

OREGANO: Simulating Circuits With The Right Seasoning

If you are looking for a simple circuit simulation software for your project, you can go for Oregano. Oregano is a graphical software application which, along with externally installed simulation engines, provides a simulation platform for circuits. Its main attractions are an easy-to-use graphical user interface and simple drag-and-drop actions.

External engines

To perform the actual simulation of circuits, you need to install another software package that is supported by Oregano. You can choose between gnuCAP/ngSpice and SPICE, all of which are open source circuit simulator engines used to check the integrity of a planned circuit design digitally.

An easy-to-use workspace

You can start off a project in Oregano from the workspace, which is a blank project page with all the actionable items and plugins available to choose from. Oregano comes with a default set of inbuilt component libraries that can be used for circuit design. Users can select from multiple libraries like the default library, TTL, linear, CPU or

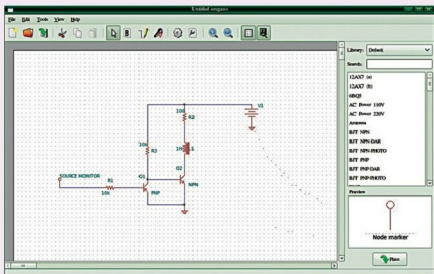


Fig. 1: The Oregano workspace

power devices library. Each library includes most of the corresponding components that are used in circuit designs. Clicking a component element gives the users a preview of the associated schematic. Elements can be included in the project by simple drag-and-drop.

Oregano makes its interface interactive and circuit development process much easier by creating special indicators for important components. For example, the software indicates ground element as GND, DC electricity lines as VDC,

while areas that may include sensors or clamps in a circuit for measurement during simulation are marked on the workspace with an indicator. All the elements are joined by lines, which indicate wire connections.

Analysing the simulation

As mentioned earlier, you have two categories of simulation analysis programs: SPICE and ngSpice/gnuCAP. If you are an Ubuntu OS user, gnuCAP will be the default analysis engine. If you prefer SPICE, you have to install it exclusively. Users can run

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